

P3↔P2 Structural Hierarchy Inheritance

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

Date: May 15, 2026

License: Creative Commons Attribution 4.0 International (CC BY 4.0)

Attribution

This work derives from and formalizes commitments across the CKS theory trilogy: “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026), and “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026).

Abstract

Paper 2 of the CKS theory trilogy establishes a three-tier structural hierarchy — cells, aspects, and Selves — each tier carrying its own DNA layer and action layer and operating over the tier below as its governance scope. Paper 3 extends this hierarchy to the inter-Self scope by choosing the aspect tier as the unit of exchange in Full Aspect Integration (FAI): when a Self contributes content to the shared substrate, it contributes aspects. This note formalizes the inheritance edge. The choice of the aspect tier as the exchange unit follows directly from the governance properties the hierarchy assigns to that tier: the aspect carries its own DNA layer and content-domain specification, making it self-contained as a governance unit; it does not expose the cell internals or the Self-level integration architecture, making it appropriately bounded for inter-organizational exchange. Paper 3 also formalizes content tier boundaries that map the three-tier hierarchy onto the shared substrate: cell-tier content enters via the aspect; aspect-tier content is contributed directly; Self-tier content stays within the contributing Self’s home governance perimeter by default. The structural hierarchy is preserved in full; the aspect tier gains an extended role as the inter-Self exchange unit. Any adversarial claim about inter-organizational AI governance content exchange must address that the exchange unit and contribution boundaries are defined by Paper 2’s structural hierarchy, not introduced fresh by Paper 3.

1. Source Commitment at Paper 2 Scope

Paper 2 (Li, April 2026, §5.2) establishes the three-tier structural hierarchy as the backbone of a CKS-governed Self. Three levels are defined, each with distinct governance scope and distinct internal structure.

The **cell tier** is the atomic coordination unit. A cell carries a DNA layer — stabilized orchestration substrates, behavior substrates, schemas, and governance rules that define how the cell functions —

and an action layer — the recorded task instances and their outputs that accumulate as the cell’s lived operational history. The cell tier handles a specific informational task. It is the atomic governance unit: self-contained, composable, and governed by the containing aspect.

The **aspect tier** is the coordination layer grouping cells around a shared content domain. An aspect carries its own DNA layer and action layer at the aspect level, coordinates its constituent cells, and operates over those cells as its content domain. Multiple aspects can coexist within one Self, potentially sharing cells across arrangements. The aspect tier is governed by the Self. Critically, the aspect carries both a domain specification — a defined purpose that gives the grouping its coherence — and governance authority over its member cells, which the cell tier alone does not carry.

The **Self tier** is the integration architecture unifying multiple aspects into one governed whole. The Self carries Self-level DNA and action layers, holds institutional knowledge as Self-level substrate content, and exercises governance authority over its aspects and, where purpose requires, directly over cells. The Self tier is the outermost governance perimeter in Paper 2. It is the level at which the full integration architecture — how the aspects relate, how cross-aspect decisions are made, what the Self’s overall strategic and operational commitments are — is held.

Each tier has its own governance authority scope. Cell governance is governed by the containing aspect. Aspect governance is governed by the Self. Self governance is governed by the humans who hold authority over the Self’s substrate content and orchestration rules.

2. Extension to Inter-Self Scope

Paper 3 (Li, April 2026) operates at the inter-Self scope: coordination across the governance perimeters of two or more distinct Selves. The central architectural move of Paper 3’s Claim 2 is to specify the unit at which this inter-Self exchange occurs. That unit is the aspect.

(a) Aspect as exchange unit (D1.06). When a Self contributes content to the shared substrate during a Full Aspect Integration event, the contribution surfaces aspects — not individual cells and not the Self-level integration architecture. The aspect is contributed as a whole: its constituent cells’ DNA-layer and action-layer content is surfaced through the aspect; the aspect’s own DNA-layer and action-layer content is directly included. The Self-level integration content remains within the contributing Self’s home governance perimeter by default, unless an explicit governance decision elevates it.

(b) Content tier boundaries (D2.49). When the three-tier hierarchy is applied to the shared-substrate contribution, three distinct contribution rules follow — one for each tier. Cell-tier content (the DNA and action layer content of the aspect’s constituent cells) enters the shared substrate via the aspect: the aspect surfaces it, the shared substrate receives it through that surface, and the cell tier does not contribute independently. Aspect-tier content (the DNA and action layer content of the aspect itself) is directly contributed to the shared substrate. Self-tier content (the DNA and action layer content held at the Self level) stays within the contributing Self’s home perimeter by default; it enters the shared substrate only when an explicit governance decision requires it.

These three rules formalize the three-tier hierarchy as the shared-substrate contribution boundary.

3. Why the Aspect Tier

The choice of the aspect tier as the inter-Self exchange unit is not an arbitrary design preference. It follows from the governance properties the three-tier hierarchy assigns to each level.

The cell tier is too granular for inter-Self exchange. Individual cells handle specific informational tasks; they carry their own DNA and action layers, but they lack the domain specification and cross-cell coordination context that make a contributed unit meaningful at inter-organizational scope. Exchanging at the cell level would require the receiving Self to reconstruct the coordination context the cell's parent aspect provides. That reconstruction would require access to the contributing Self's aspect-level governance information, which is not what was shared.

The Self tier is too comprehensive for inter-Self exchange. The Self holds the full integration architecture — how aspects relate, how cross-aspect decisions are made, what the Self's overall commitments are. Contributing at the Self level would expose this integration architecture in its entirety to participating organizations. In most inter-organizational contexts, that exposure would be neither appropriate nor intended. The Self-level architecture is the contributing organization's governance design; it is not the domain-specific coordination content the exchange is meant to share.

The aspect tier occupies the governance boundary that makes inter-Self exchange coherent. An aspect carries a domain specification — a defined purpose that gives the grouping its meaning — and it carries governance authority over its constituent cells. It is self-contained as a governance unit: a recipient Self can receive a contributed aspect and understand what it is for, what rules govern its cells, and what content it carries, without needing access to either the cell internals in isolation or the contributing Self's integration architecture. The aspect is the level of governance granularity at which content is both complete enough to be useful and bounded enough to be shared across organizational governance perimeters.

4. What Is Preserved; What Is Extended

The three-tier structural hierarchy from Paper 2 is preserved in full across the extension to Paper 3. Cells remain the atomic coordination units with their own DNA and action layers. Aspects remain coordination arrangements of cells with their own DNA and action layers and their own governance scope over member cells. Selves remain the integration architectures holding multiple aspects under unified governance. The governance authority relationships — cell governed by aspect, aspect governed by Self — are unchanged. The DNA/action layer distinction within every tier carries forward without modification.

What is extended is the aspect tier's role. In Paper 2, aspects are intra-Self organizational units: they exist within a Self's home governance perimeter and organize cells toward a shared purpose within that perimeter. Paper 3 assigns the aspect tier a new role at the inter-Self boundary: the aspect is the

unit of exchange in FAI, the unit at which a Self's content crosses the boundary into the shared substrate. The aspect moves from being a purely intra-Self organizational structure to being simultaneously an intra-Self organizational structure and an inter-Self exchange unit.

This role extension leaves the hierarchy's definition intact. The aspect is the same architectural object — a governed coordination arrangement of cells with its own DNA and action layers — performing the same functions within each Self's home perimeter and additionally performing the exchange function at the inter-Self boundary. The extension is a new deployment of the existing structure, not a redefinition of it.

5. Prior-Art Inheritance Claim

Paper 3's aspect-as-exchange-unit commitment (D1.06) and content tier boundaries (D2.49) directly inherit from Paper 2's three-tier structural hierarchy (B0.04). The inheritance operates on two levels.

First, the identification of the aspect tier as the exchange unit is a consequence of the governance properties the hierarchy defines. Any system that implements inter-Self coordination using an aspect-level exchange unit is using Paper 2's three-tier hierarchy's governance structure to define that exchange boundary. The choice is not independent of the hierarchy; it follows from the hierarchy.

Second, the content tier boundaries — cell-tier content via aspect, aspect-tier content directly, Self-tier content home by default — are a direct mapping of the three-tier hierarchy onto the shared-substrate contribution rule. Any system that implements these contribution boundaries is applying Paper 2's three-tier structure to the inter-Self contribution context.

Any adversarial claim that the aspect-as-exchange-unit is a novel invention independent of prior work must address that the exchange unit is fully determined by Paper 2's hierarchy: the aspect tier is the level of governance granularity that is self-contained without being over-comprehensive, and that determination is Paper 2's, not Paper 3's. Paper 3's contribution is the role extension — the application of an existing structure to a new scope — not the structural hierarchy on which the extension depends.

Operational Test

A system instantiates the Paper 3 structural hierarchy inheritance if all of the following hold:

1. When a governed AI entity contributes content to an inter-organizational shared substrate, it contributes at the aspect level — not at the level of individual cells and not at the level of its full integration architecture.
2. The contributed aspect surfaces the DNA-layer and action-layer content of its constituent cells without those cells contributing independently.
3. The aspect's own DNA-layer and action-layer content is directly present in the shared substrate.

4. Content held at the entity's integration layer — the coordination rules and commitments that govern how its aspects relate — remains within the entity's home governance perimeter by default, absent an explicit governance decision to include it.

5. The unit of exchange (the aspect) carries a content-domain specification that makes it interpretable as a governance unit without reference to the contributing entity's integration architecture.

A system that exchanges at cell granularity, at full-entity granularity, or at a granularity that does not correspond to the three-tier hierarchy's aspect level has departed from this inheritance and cannot claim the prior-art defense the hierarchy provides.

Source Papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

How to Cite This Note

Li, W. (2026). *P3↔P2 Structural Hierarchy Inheritance*. May 15, 2026. ORCID: 0009-0004-8065-3235.